

DEV AGRAWAL

B.Tech Computer Science and Engineering — VIT Bhopal University — Expected Graduation: 2028

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*Software Engineering student and full-stack developer specializing in graphics programming, computer vision, and AI systems.
Builds scalable, production-grade software spanning real-time WebGL rendering, distributed backend services, and
knowledge-graph-driven applications.*

Education

Vellore Institute of Technology, Bhopal

Expected May 2028

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.26/10

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML5, CSS3

Frontend: React, Vite, Tailwind CSS, Three.js, React Three Fiber, Phaser

Backend: FastAPI, REST APIs, Node.js, Microservices, WebSockets

Graphics: WebGL, Three.js, Real-Time Rendering, Physics Simulation, Procedural Generation

AI: Computer Vision (MediaPipe), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LiteLLM, Ollama

Databases: PostgreSQL, Neo4j (Graph Database), SQL

DevOps: Docker, Kubernetes, OpenShift, CI/CD, Linux, Bash

Developer Tools: Git, GitHub, VS Code, Postman

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Software Engineering

Projects

Interactive Intelligence Platform

Live Demo — GitHub

- Architected a persistent, browser-native WebGL platform in **React, TypeScript, Three.js, and React Three Fiber**, unifying computer vision, physics simulation, procedural generation, and graphics programming across 10+ interactive experiences via a shared, reusable rendering architecture with zero page reloads.
- Engineered **Gesture Lab**, a **MediaPipe**-powered computer vision suite delivering real-time hand-tracking across 8 modules – particle systems, smoke and rigid-body physics, Boids flocking, reaction-diffusion, neural visualization, holographic rendering, and freehand drawing – each optimized for smooth browser-native execution.
- Built **Computational Arcade**, implementing 5 pathfinding and search algorithms (BFS, DFS, Dijkstra, A*, Greedy Best-First Search) with live visualization in Maze Explorer, alongside Space Shooter and Sector Runner sharing a unified leaderboard service and an optimized, reusable rendering pipeline.

Neural Nexus

GitHub

- Designed a full-stack AI memory engine that converts conversations, documents, and project data into a continuously evolving **knowledge graph**, solving context loss across multi-session LLM workflows; built with **Python, FastAPI, React, PostgreSQL, and Neo4j**.
- Implemented context-aware, **Retrieval-Augmented Generation (RAG)** by combining **Graphiti**-based temporal graph relationships with **LiteLLM** and **Ollama** for multi-model orchestration across local and cloud language models.
- Developed **REST APIs** for semantic memory ingestion and graph relationship management, paired with an interactive React dashboard for knowledge graph visualization, activity timeline, and memory analytics – containerized end-to-end with **Docker** for one-command deployment.

Certifications

IBM DevOps and Software Engineering Professional Certificate (15-course Professional Certificate)

Google IT Support Professional Certificate (6-course Professional Certificate)

Relevant Technologies: Git & GitHub, Linux Shell Scripting, Containers (Docker, Kubernetes & OpenShift), Microservices & Serverless Architecture, CI/CD, Test & Behavior Driven Development, Application Security for Developers